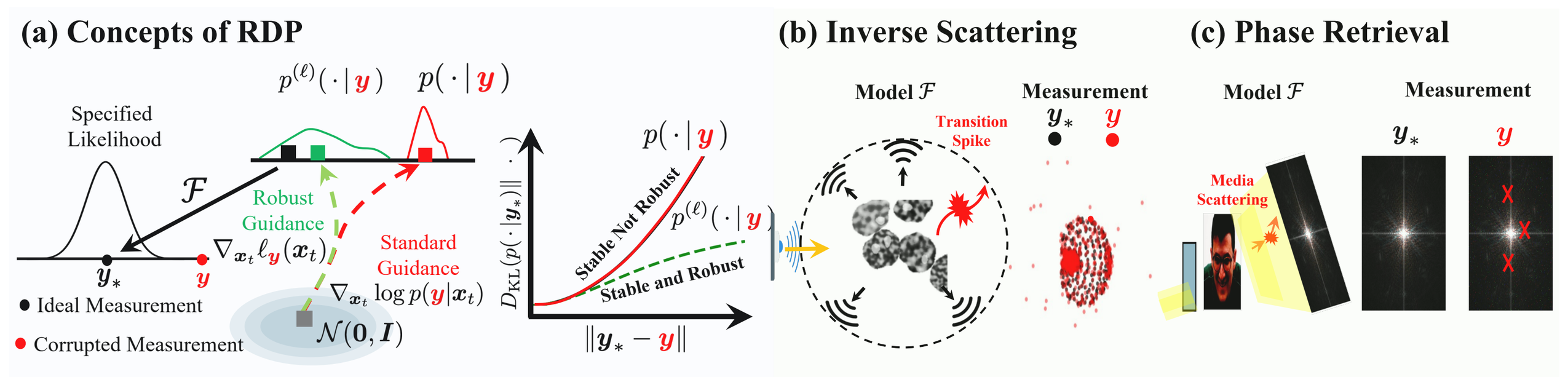
(a) Concepts of RDP
$p^{(\ell)}(\cdot\,|\,\boldsymbol{y})$
$p(\cdot\,|\,\boldsymbol{y})$
Specified Likelihood
$\mathcal{F}$
Robust Guidance
$\nabla_{\boldsymbol{x}_t}\ell_{\boldsymbol{y}}(\boldsymbol{x}_t)$
Standard Guidance
$\nabla_{\boldsymbol{x}_t}\log p(\boldsymbol{y}|\boldsymbol{x}_t)$
$\mathcal{N}(\mathbf{0},\boldsymbol{I})$
$\boldsymbol{y}_*$
$\boldsymbol{y}$
Ideal Measurement
Corrupted Measurement
$D_{\mathrm{KL}}(p(\cdot\,|\,\boldsymbol{y}_*)\|\,\cdot\,)$
$p(\cdot\,|\,\boldsymbol{y})$
Stable Not Robust
$p^{(\ell)}(\cdot\,|\,\boldsymbol{y})$
Stable and Robust
$\|\boldsymbol{y}_*-\boldsymbol{y}\|$
(b) Inverse Scattering
Model $\mathcal{F}$
Measurement
Transition Spike
$\boldsymbol{y}_*$
$\boldsymbol{y}$
(c) Phase Retrieval
Model $\mathcal{F}$
Measurement
Media Scattering
$\boldsymbol{y}_*$
$\boldsymbol{y}$